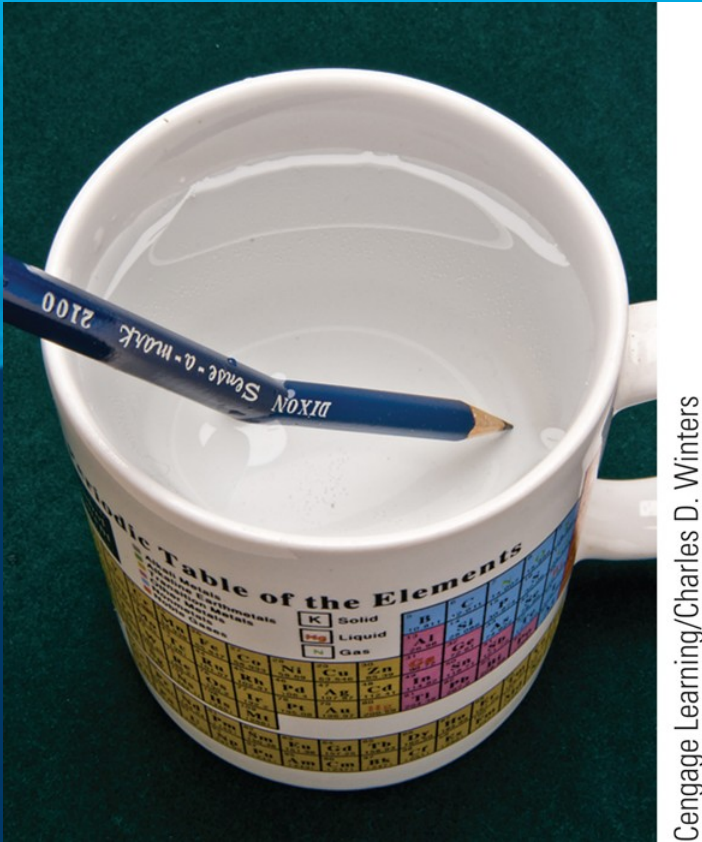




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Topic 22

Reflection and Refraction of Light

Reading Question 22-1 (1 of 2)

The theory proposing light as a particle is supported by

1. the sparks created in Hertz' experiment with the spark gap oscillator.
2. the speed of light turning up in Maxwell's equations.
3. the interference caused by light in Young's double- slit experiment.
4. the photoelectric effect Hertz discovered.

Reading Question 22-1 (2 of 2)

The theory proposing light as a particle is supported by

1. the sparks created in Hertz' experiment with the spark gap oscillator.
2. the speed of light turning up in Maxwell's equations.
3. the interference caused by light in Young's double- slit experiment.
4. **Answer: the photoelectric effect Hertz discovered.**

Reading Question 22-2 (1 of 2)

A ray of light hits a flat surface at an angle θ and is diffusely reflected. At what angle is it reflected?

1. θ
2. 2θ
3. $-\theta$
4. unknown

Reading Question 22-2 (2 of 2)

A ray of light hits a flat surface at an angle θ and is diffusely reflected. At what angle is it reflected?

1. θ
2. 2θ
3. $-\theta$
4. **Answer: unknown**

Reading Question 22-3 (1 of 2)

When a beam of white light is refracted through a prism, which of the following colors has the greatest angular deviation from the original direction of the beam?

1. red
2. yellow
3. blue
4. violet

Reading Question 22-3 (2 of 2)

When a beam of white light is refracted through a prism, which of the following colors has the greatest angular deviation from the original direction of the beam?

1. red
2. yellow
3. blue
4. **Answer: violet**

Reading Question 22-4 (1 of 2)

Light passes from a material with index of refraction 1.1 into one with index of refraction 1.5 after being incident at a non-normal angle ($\theta_{\text{inc}} \neq 0$). Compared to the incident ray, the refracted ray will

1. be unaffected.
2. bend towards the normal.
3. follow the normal.
4. bend away from the normal.

Reading Question 22-4 (2 of 2)

Light passes from a material with index of refraction 1.1 into one with index of refraction 1.5 after being incident at a non-normal angle ($\theta_{\text{inc}} \neq 0$). Compared to the incident ray, the refracted ray will

1. be unaffected.
- 2. Answer: bend towards the normal.**
3. follow the normal.
4. bend away from the normal.

Reading Question 22-5 (1 of 2)

Huygen's principle states that

1. there is a secondary rainbow with the spectral colors reversed from the primary rainbow.
2. light has properties of both waves and particles.
3. all points on a wave front can be taken as point sources of secondary spherical wavelets which construct the new position of the wave front as time passes.
4. rainbows are a result of dispersion.

Reading Question 22-5 (2 of 2)

Huygen's principle states that

1. there is a secondary rainbow with the spectral colors reversed from the primary rainbow.
2. light has properties of both waves and particles.
3. **Answer: all points on a wave front can be taken as point sources of secondary spherical wavelets which construct the new position of the wave front as time passes.**
4. rainbows are a result of dispersion.

Topic 22: Reflection and Refraction of Light

TOPIC DISCUSSION

The Nature of Light

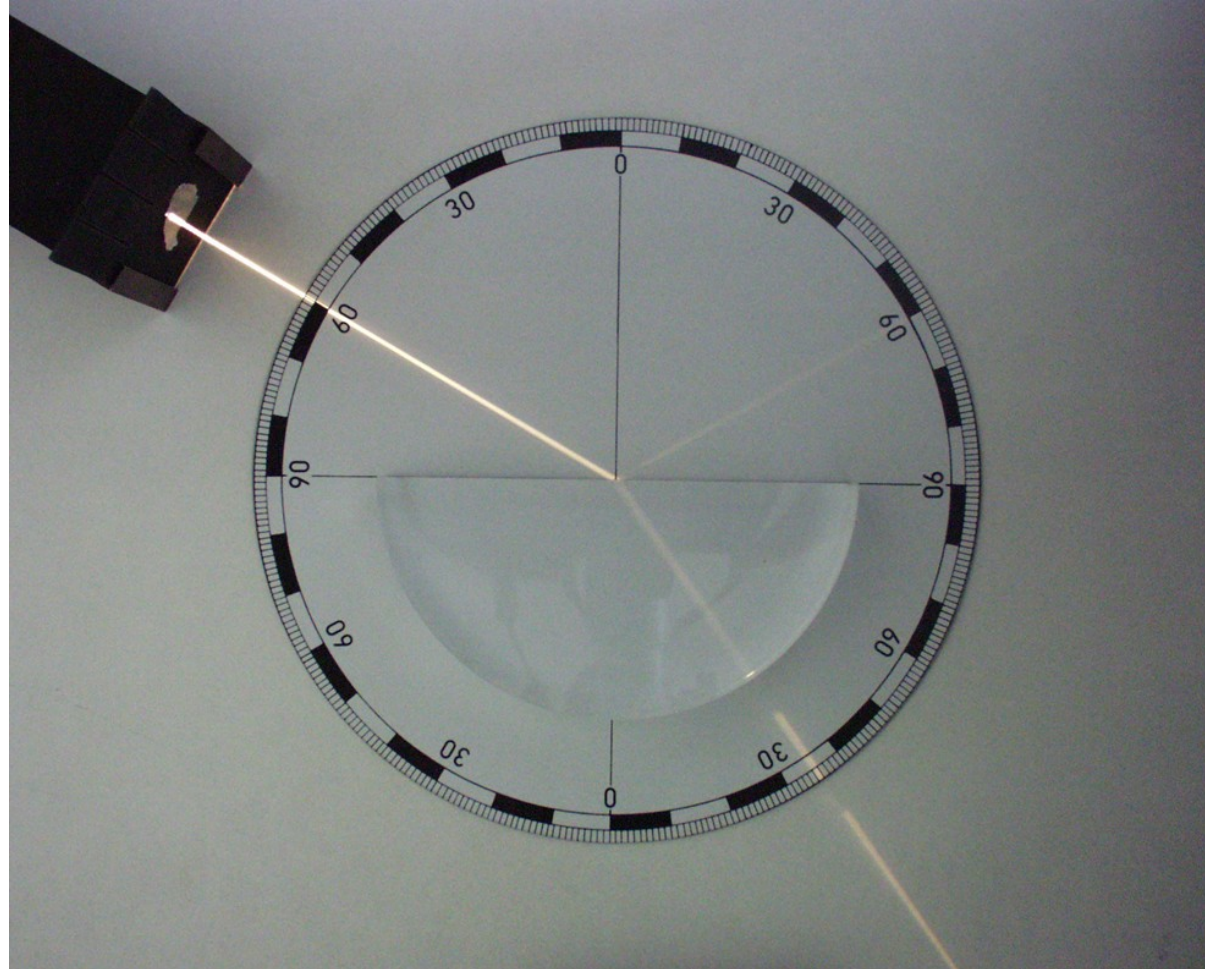


$$E = hf$$

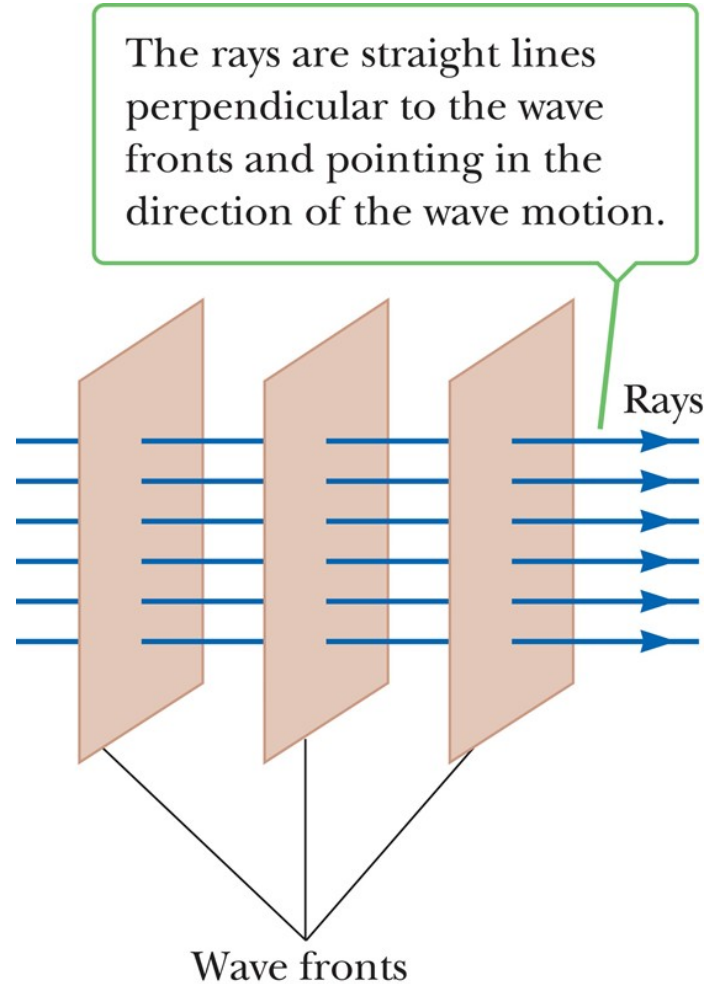
$$h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$$

Dual Nature of Light:
In some experiments
light acts as a wave
and in others it acts
as a particle.

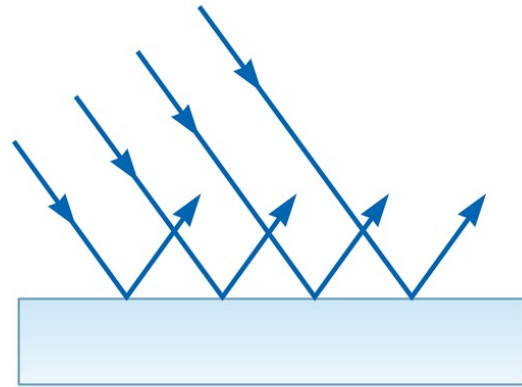
Reflection and Refraction



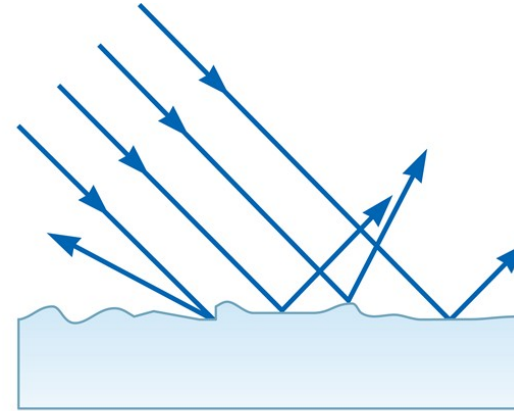
The Ray Approximation in Geometric Optics



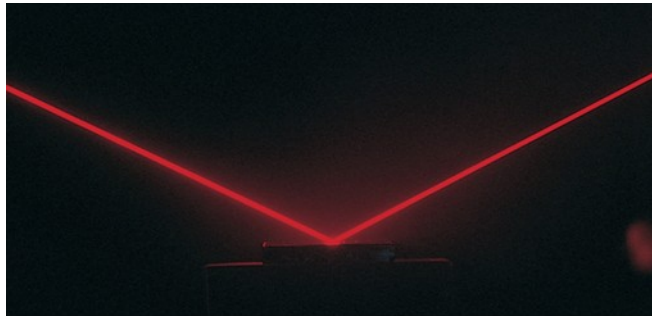
Reflection of Light (1 of 2)



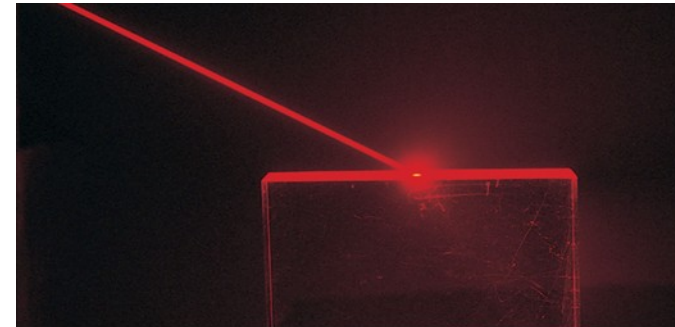
a



b



c



d

Photographs Courtesy of Henry
Leap and Jim Lehman

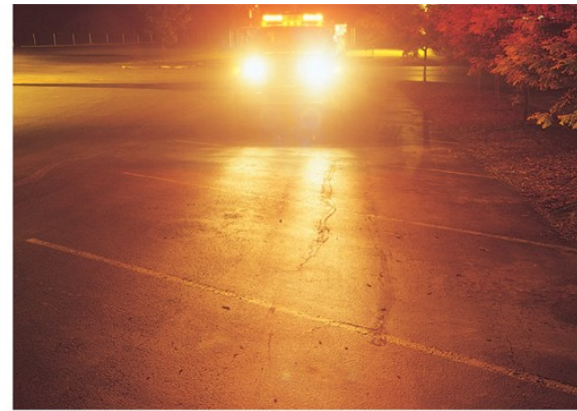
Think – Pair – Share 1

Which part of the figure below, (a) or (b), better shows specular reflection of light from the roadway?

1. (a)
2. (b)
3. Neither show specular reflection.



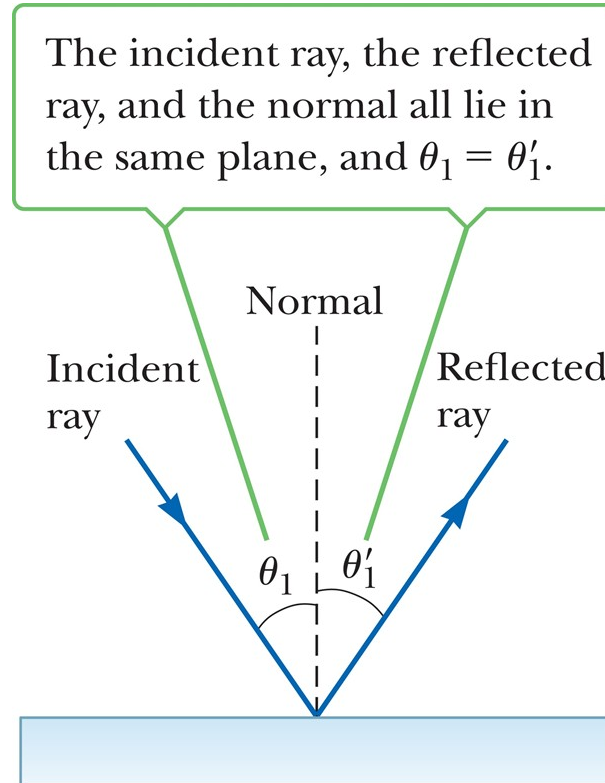
a



b

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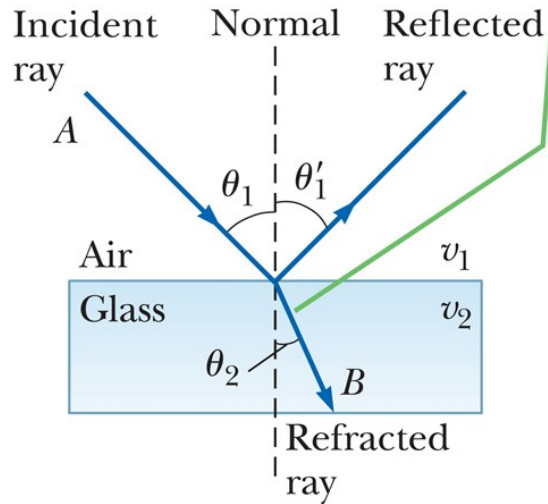
Reflection of Light (2 of 2)



angle of incidence = angle of reflection

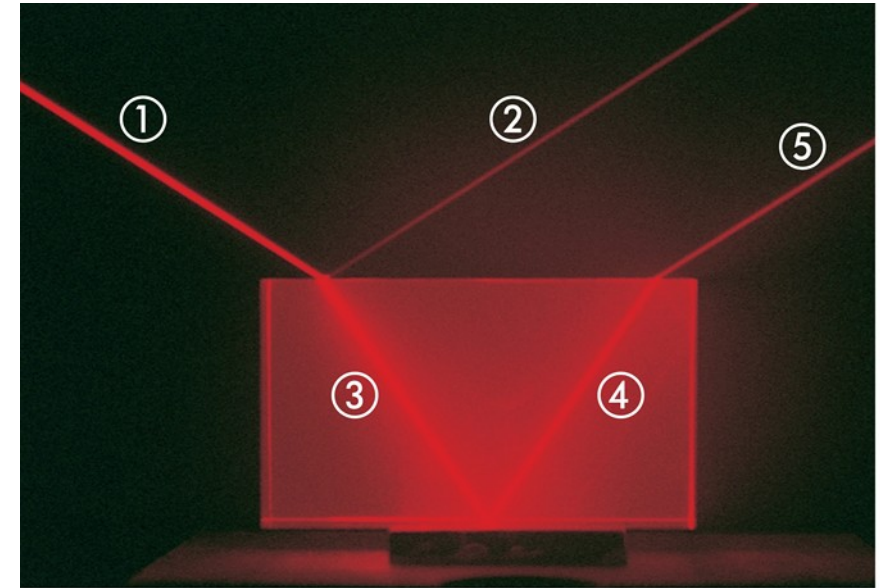
Refraction of Light (1 of 2)

All rays and the normal lie in the same plane, and the refracted ray is bent toward the normal because $v_2 < v_1$.



a

$$\frac{\sin \theta_2}{\sin \theta_1} = \frac{v_2}{v_1} = \text{constant}$$



b

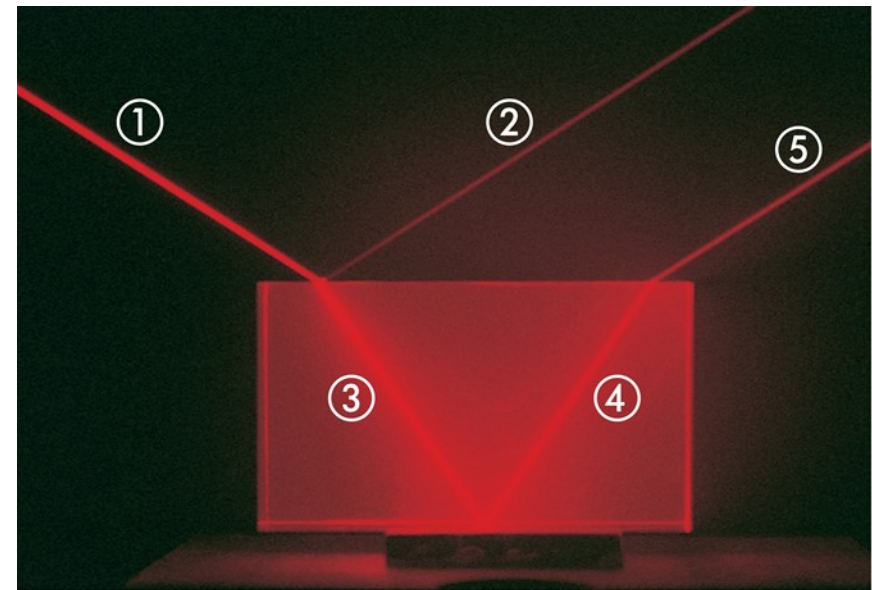
Courtesy of Henry Leap and Jim Lehman

Path of light ray through refracting surface is reversible.

Think – Pair – Share 2

If beam 1 is the incoming beam in the figure below, which of the other four beams are due to reflection?

1. beams 2 and 3
2. beam 3 and 5
3. beam 2 and 4
4. beam 5 only



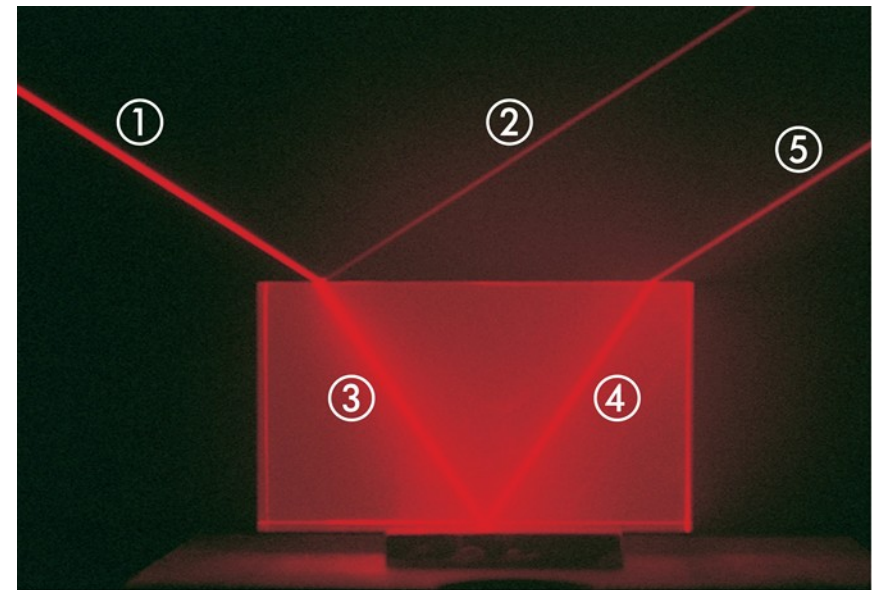
Courtesy of Henry Leap and Jim Lehman

b

Think – Pair – Share 3

If beam 1 is the incoming beam in the figure below, which of the other four beams are due to refraction?

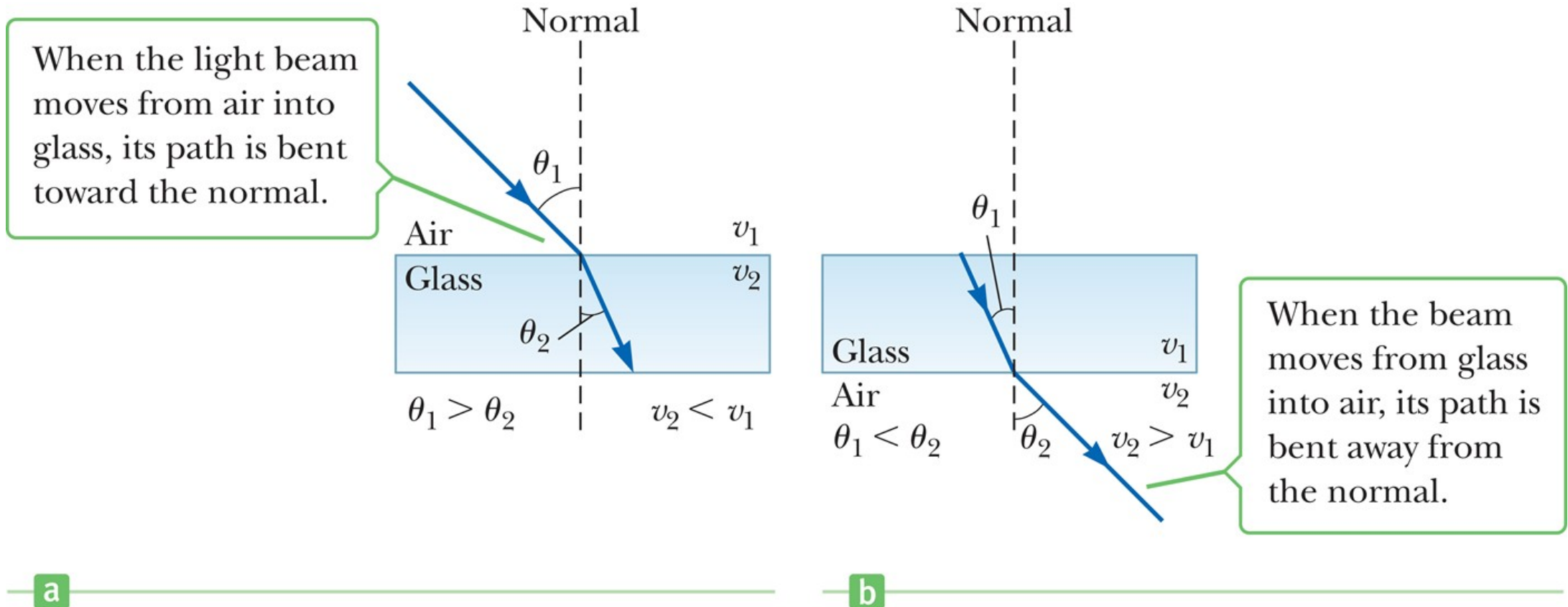
1. beams 2 and 3
2. beam 3 and 5
3. beam 2 and 4
4. beam 5 only



Courtesy of Henry Leap and Jim Lehman

b

Refraction of Light (2 of 2)



The Law of Refraction (1 of 4)

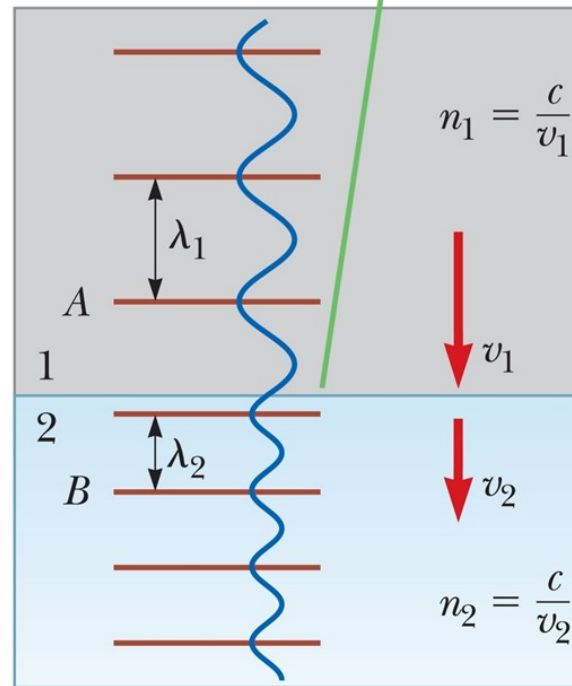
$$n \equiv \frac{\text{speed of light in vacuum}}{\text{speed of light in a medium}} = \frac{c}{v}$$

Table 22.1 Indices of Refraction for Various Substances, Measured with Light of Vacuum Wavelength $\lambda_0 = 589 \text{ nm}$

Substance	Index of Refraction	Substance	Index of Refraction
Solids at 20°C		Liquids at 20°C	
Diamond (C)	2.419	Benzene	1.501
Fluorite (CaF ₂)	1.434	Carbon disulfide	1.628
Fused quartz (SiO ₂)	1.458	Carbon tetrachloride	1.461
Glass, crown	1.52	Ethyl alcohol	1.361
Glass, flint	1.66	Glycerine	1.473
Ice (H ₂ O) (at 0°C)	1.309	Water	1.333
Polystyrene	1.49	Gases at 0°C, 1 atm	
Sodium chloride (NaCl)	1.544	Air	1.000 293
Zircon	1.923	Carbon dioxide	1.000 45

The Law of Refraction (2 of 4)

As a wave moves from medium 1 to medium 2, its wavelength changes but its frequency remains constant.



The Law of Refraction (3 of 4)

$$v = f\lambda \quad \text{and} \quad f_1 = f_2 = f$$

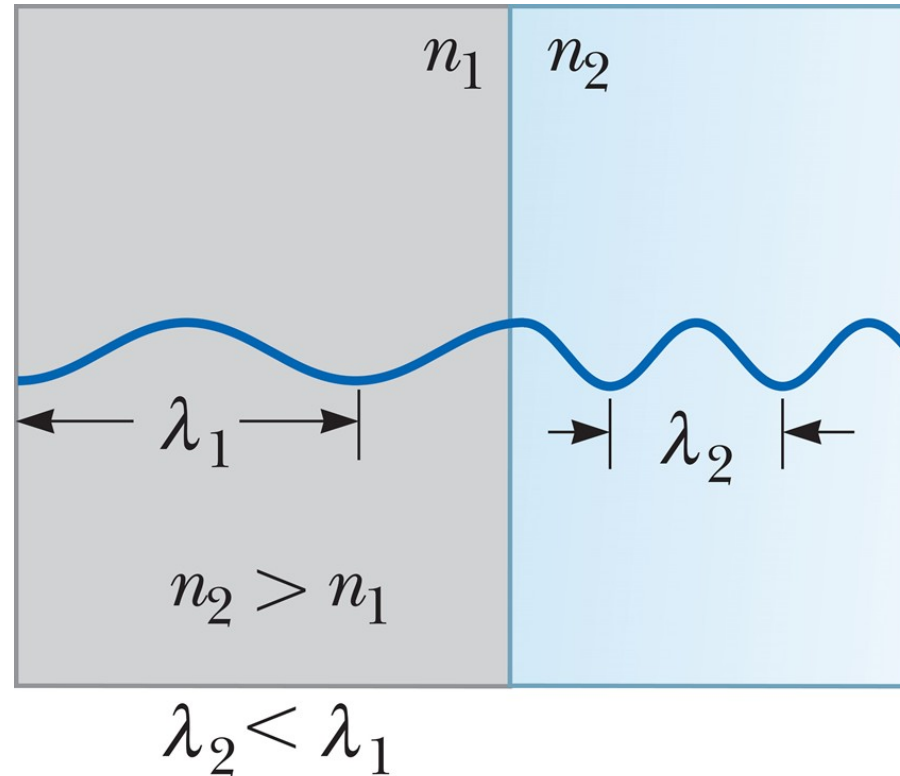
$$v_1 = f\lambda_1 \quad \text{and} \quad v_2 = f\lambda_2$$

$$\frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} \quad \rightarrow \quad v = \frac{c}{n} \quad \rightarrow \quad \frac{\lambda_1}{\lambda_2} = \frac{c/n_1}{c/n_2} = \frac{n_2}{n_1}$$

$$\lambda_1 n_1 = \lambda_2 n_2$$

$$n = \frac{\lambda_0}{\lambda_n}$$

The Law of Refraction (4 of 4)

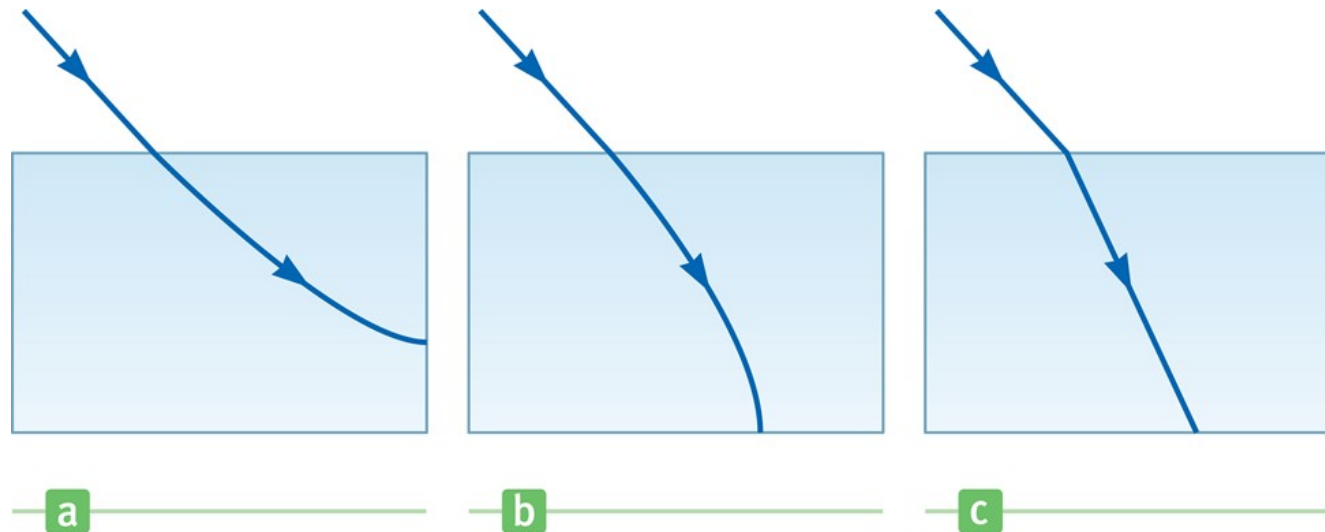


$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Think – Pair – Share 4

A material has an index of refraction that increases continuously from top to bottom. Of the three paths shown in the figure, which path will a light ray follow as it passes through the material?

1. Path (a)
2. Path (b)
3. Path (c)



Think – Pair – Share 5

As light travels from vacuum ($n = 1$) to a medium such as glass ($n > 1$), which of the following properties remains the same?

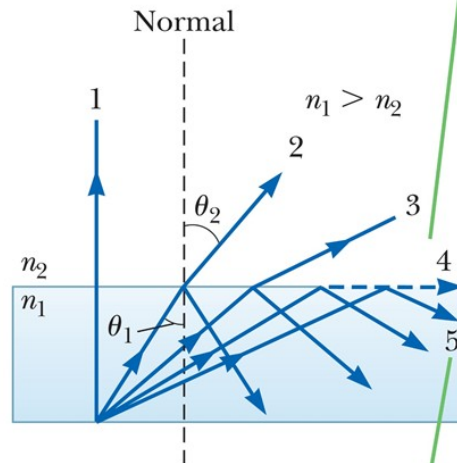
- (a) wavelength
- (b) wave speed
- (c) frequency

1. (a)
2. (b)
3. (c)
4. (b) and (c)

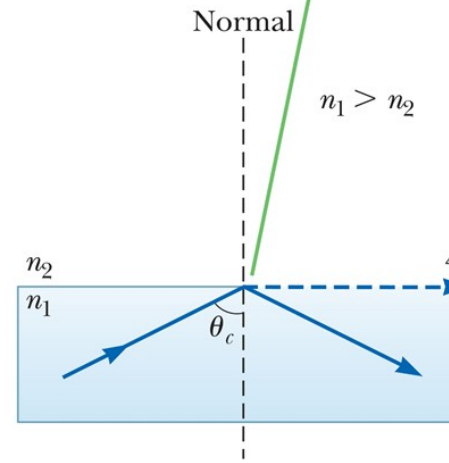
Total Internal Reflection (1 of 4)

As the angle of incidence θ_1 increases, the angle of refraction θ_2 increases until θ_2 is 90° (ray 4). The dashed line indicates that no energy actually propagates in this direction.

The angle of incidence producing an angle of refraction equal to 90° is the *critical angle* θ_c . At this angle of incidence, all the energy of the incident light is reflected.



For even larger angles of incidence, total internal reflection occurs (ray 5).



a

b

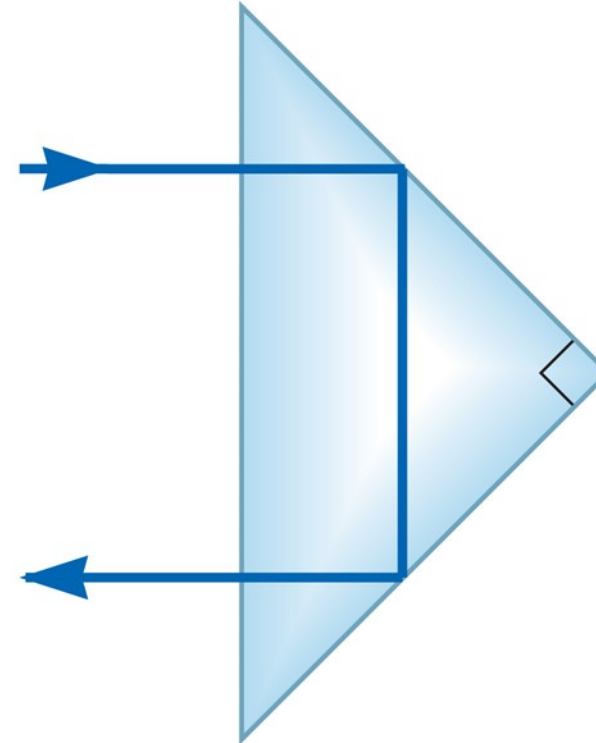
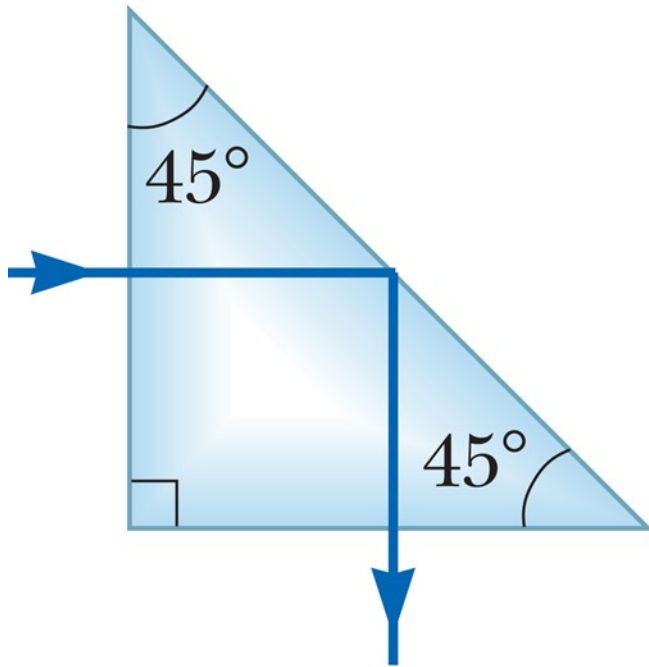
Total Internal Reflection (2 of 4)

$$n_1 \sin \theta_c = n_2 \sin 90^\circ = n_2$$

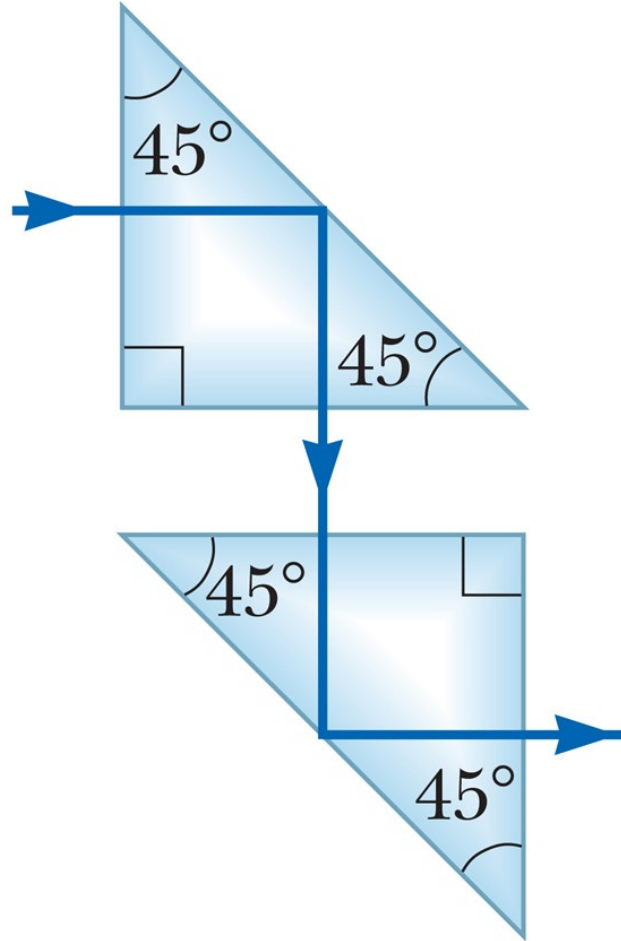
$$\sin \theta_c = \frac{n_2}{n_1} \quad \text{for } n_1 > n_2$$



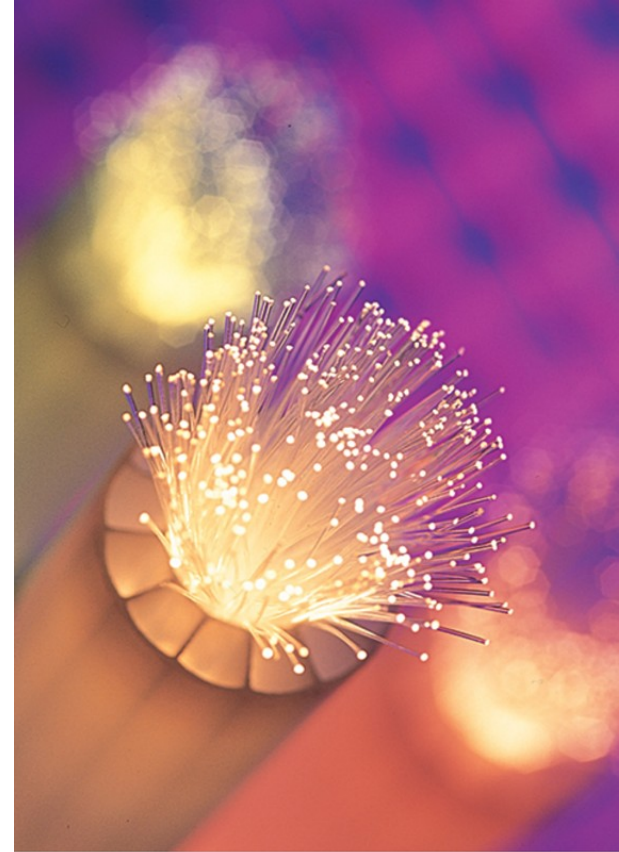
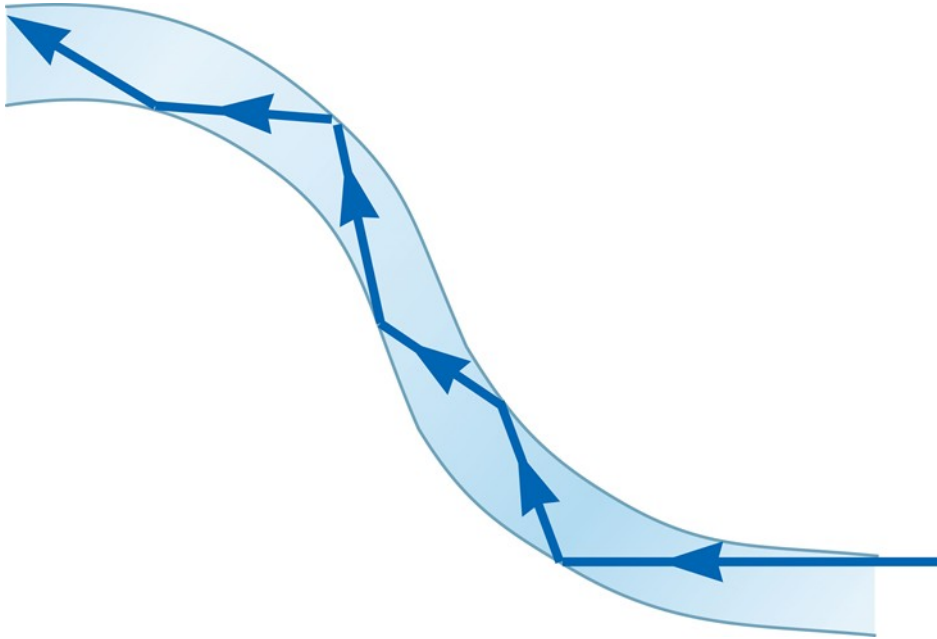
Total Internal Reflection (3 of 4)



Total Internal Reflection (4 of 4)



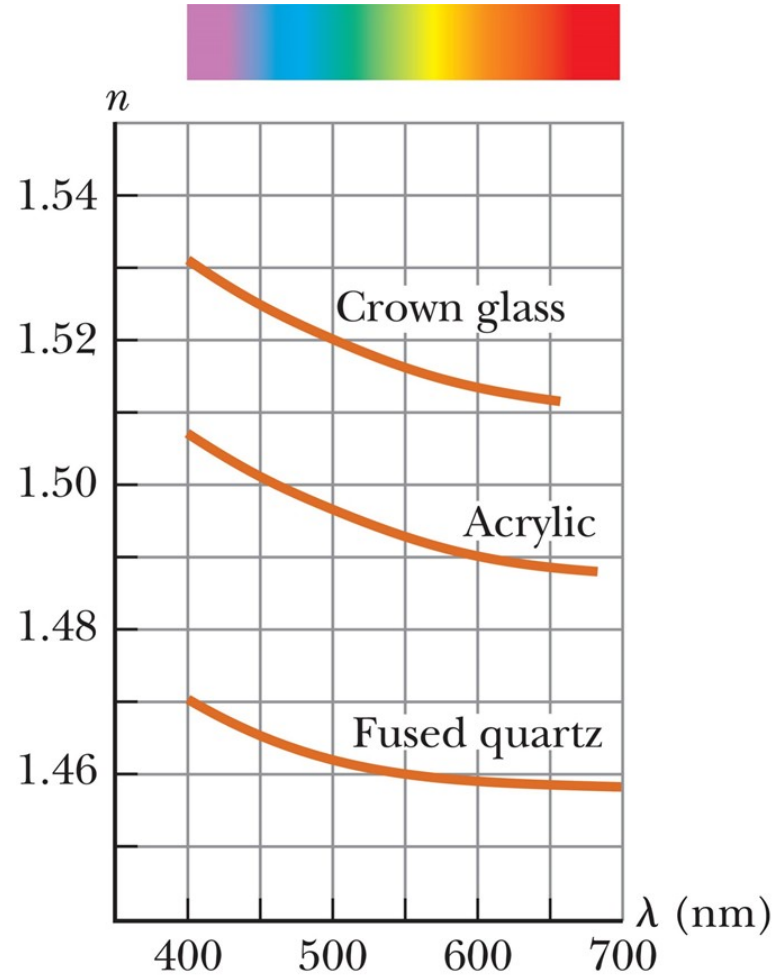
Fiber Optics



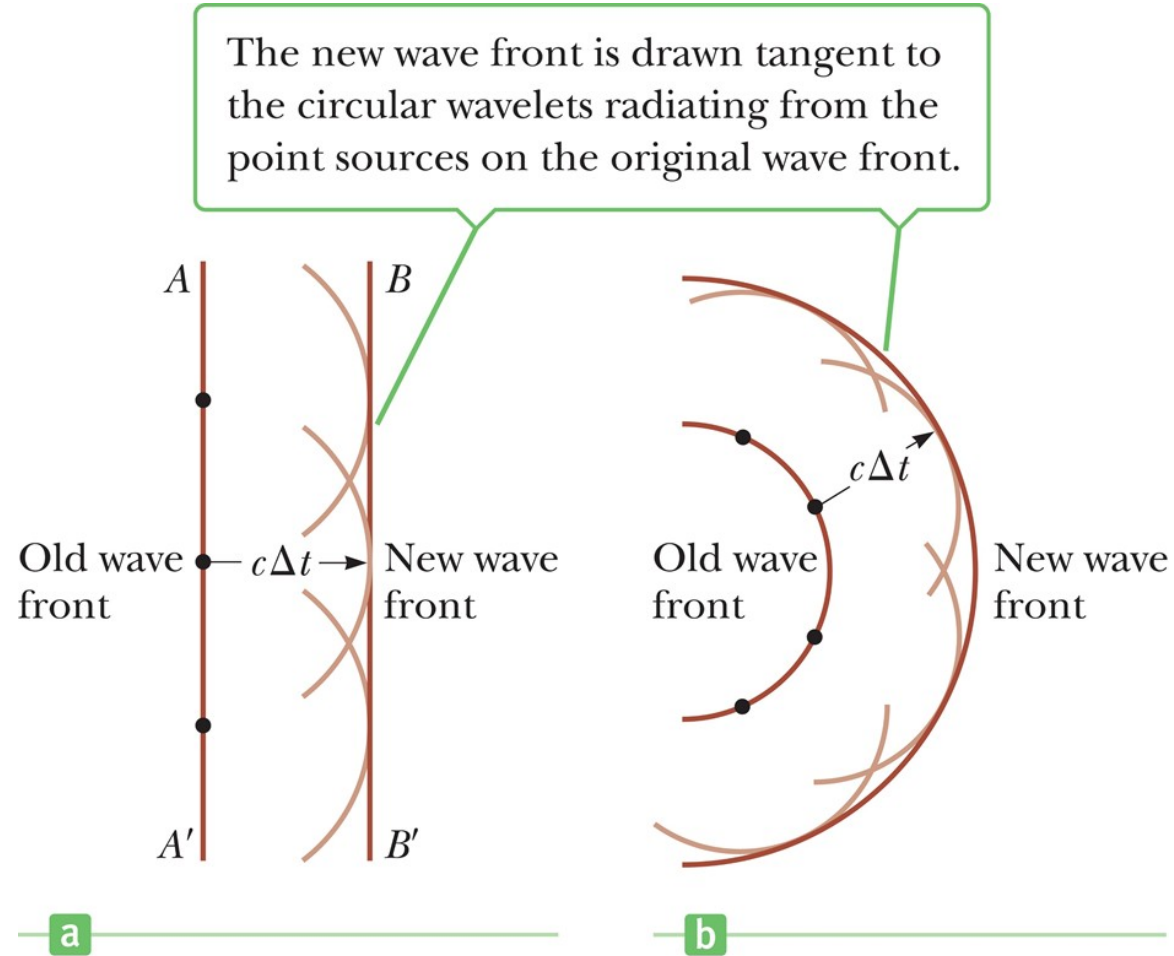
Dennis O'Clair/Tony Stone Images/Getty Images

a

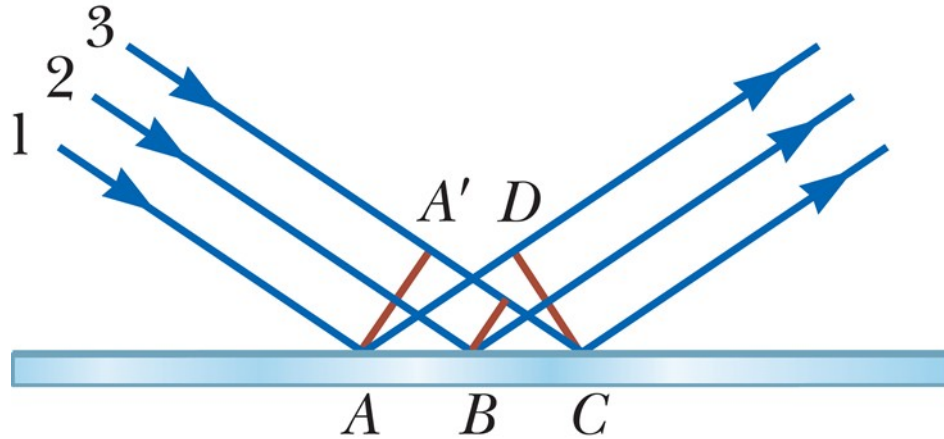
Dispersion and Prisms (1 of 5)



Huygens' Principle



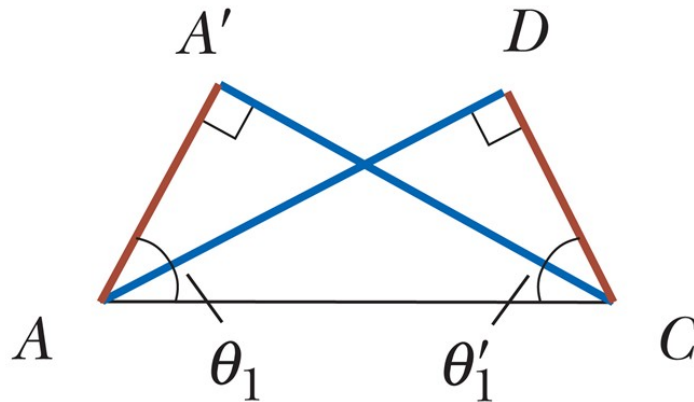
Huygens' Principle Applied to Reflection and Refraction (1 of 3)



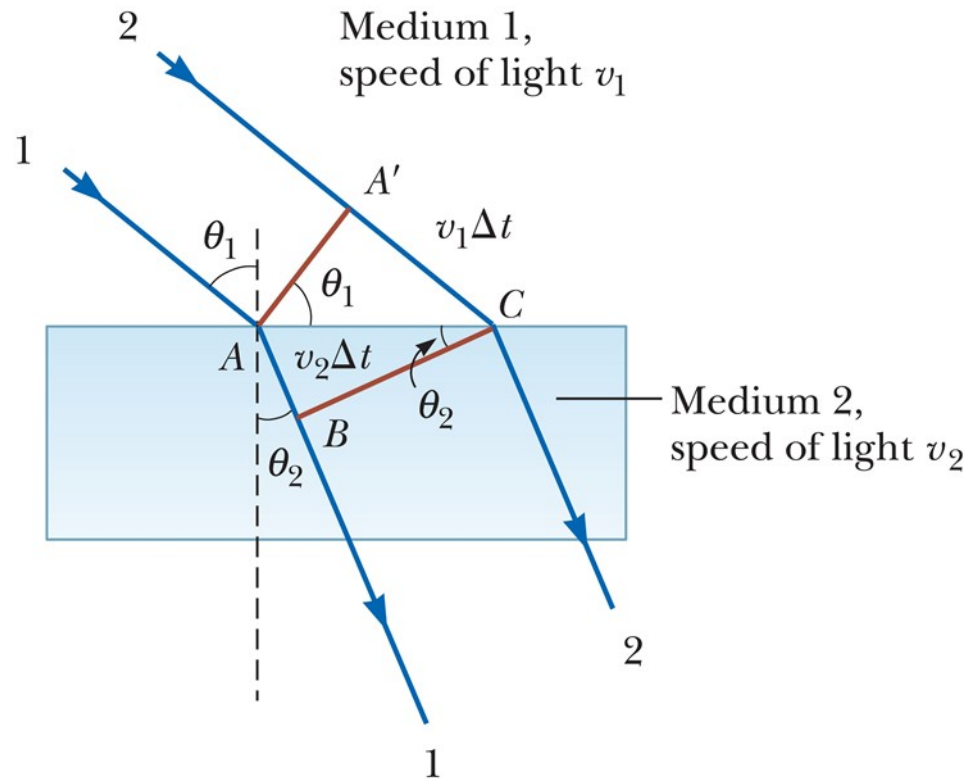
$$\sin \theta_1 = \frac{A'C}{AC} \quad \text{and} \quad \sin \theta'_1 = \frac{AD}{AC}$$

$$\sin \theta_1 = \sin \theta'_1$$

$$\theta_1 = \theta'_1$$



Huygens' Principle Applied to Reflection and Refraction (2 of 3)

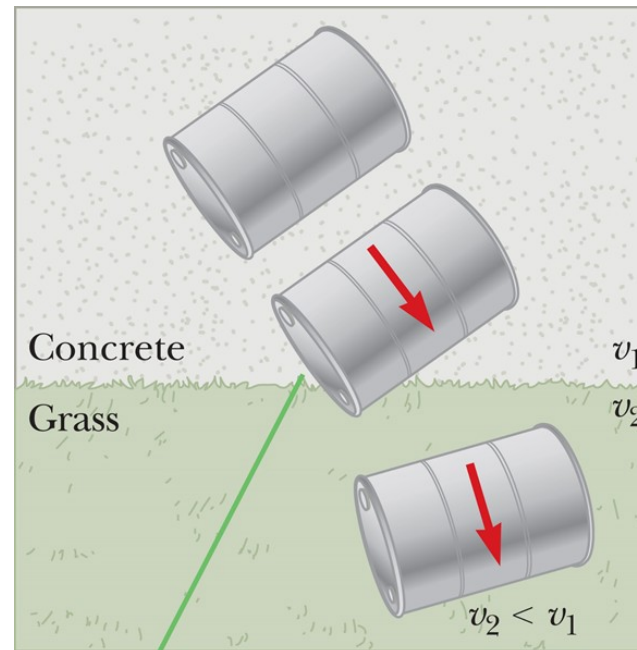


$$\sin \theta_1 = \frac{v_1 \Delta t}{AC} \quad \text{and} \quad \sin \theta_2 = \frac{v_2 \Delta t}{AC}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{c/n_1}{c/n_2} = \frac{n_2}{n_1}$$

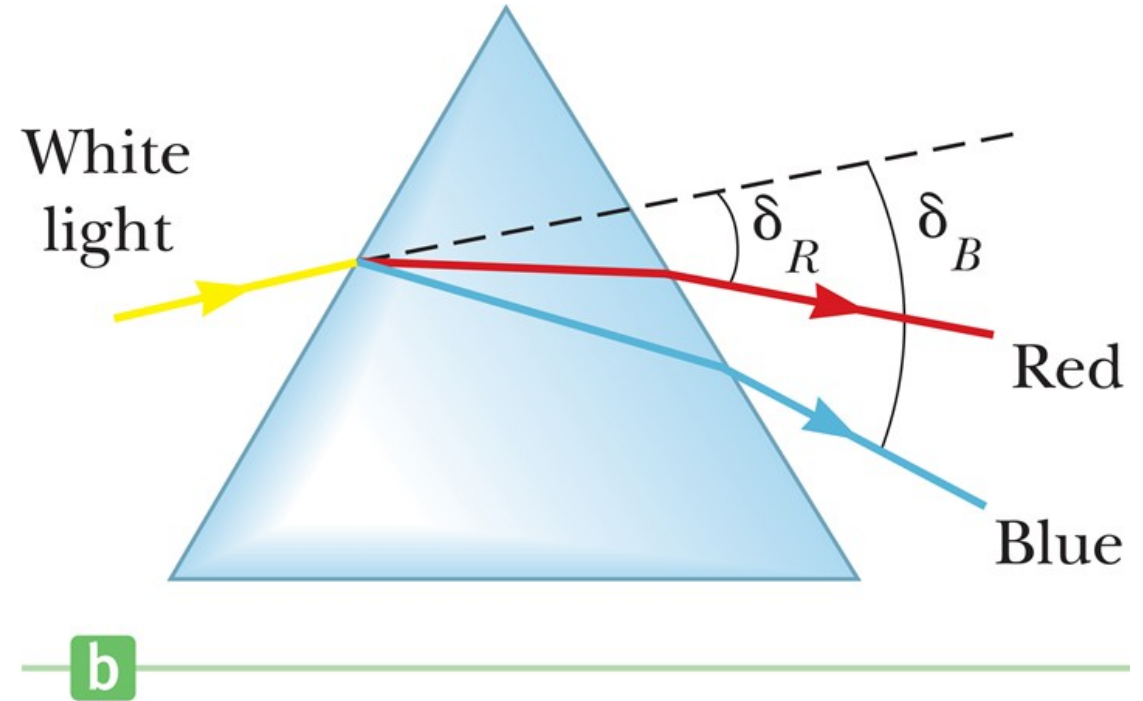
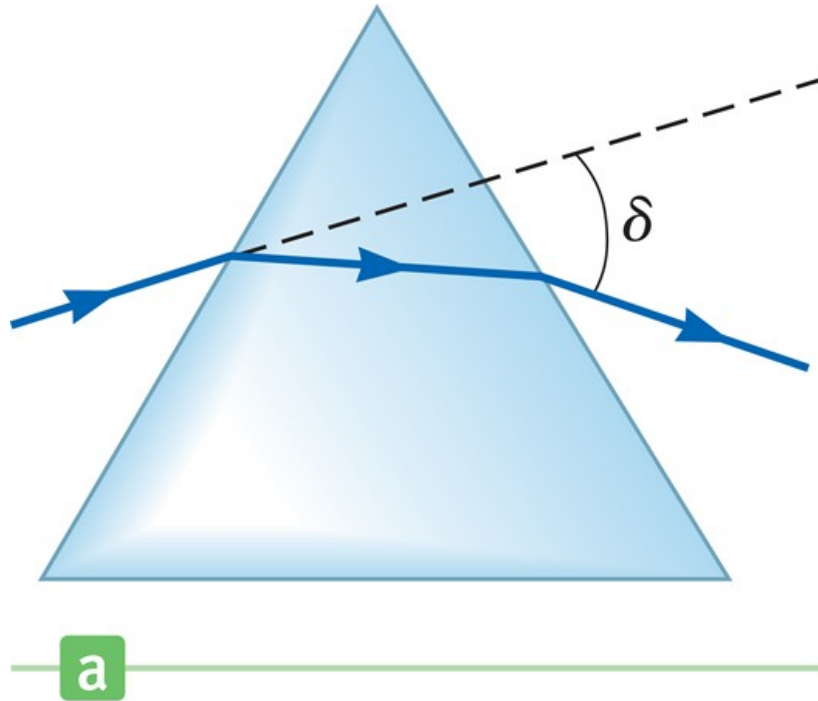
Huygens' Principle Applied to Reflection and Refraction (3 of 3)



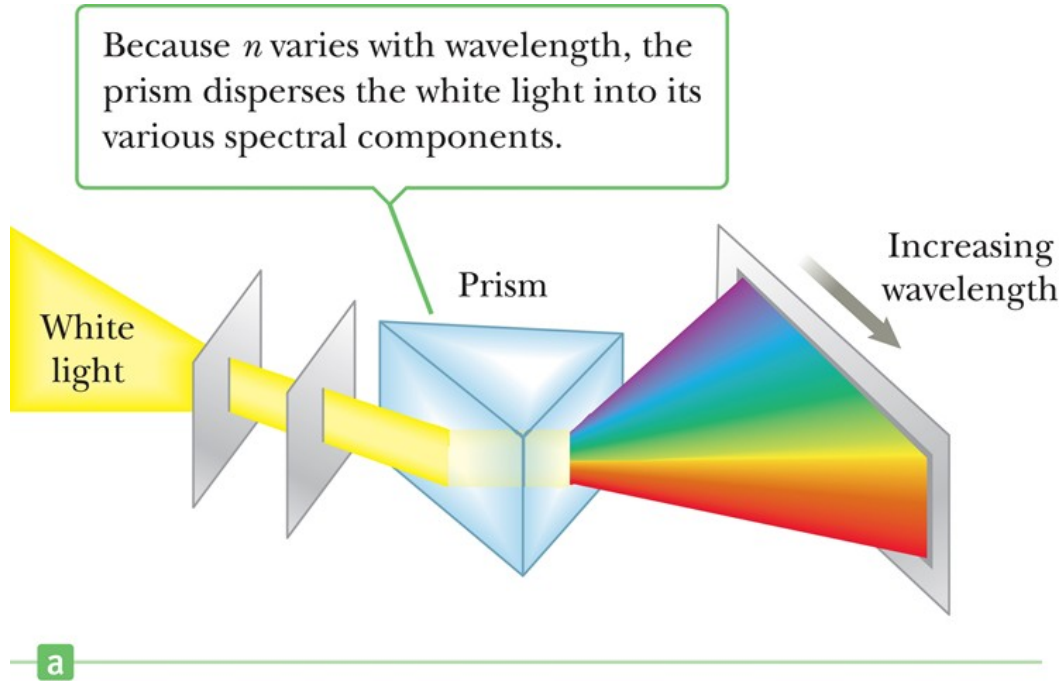
This end slows first; as a result, the barrel turns.

b

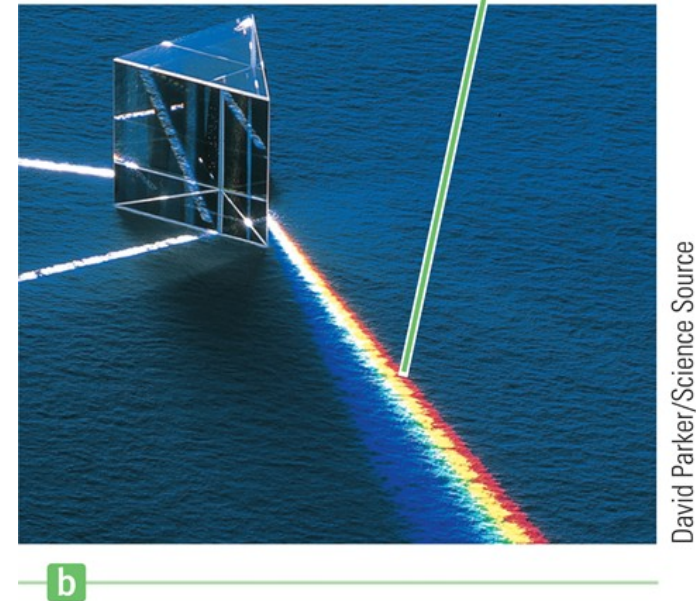
Dispersion and Prisms (2 of 5)



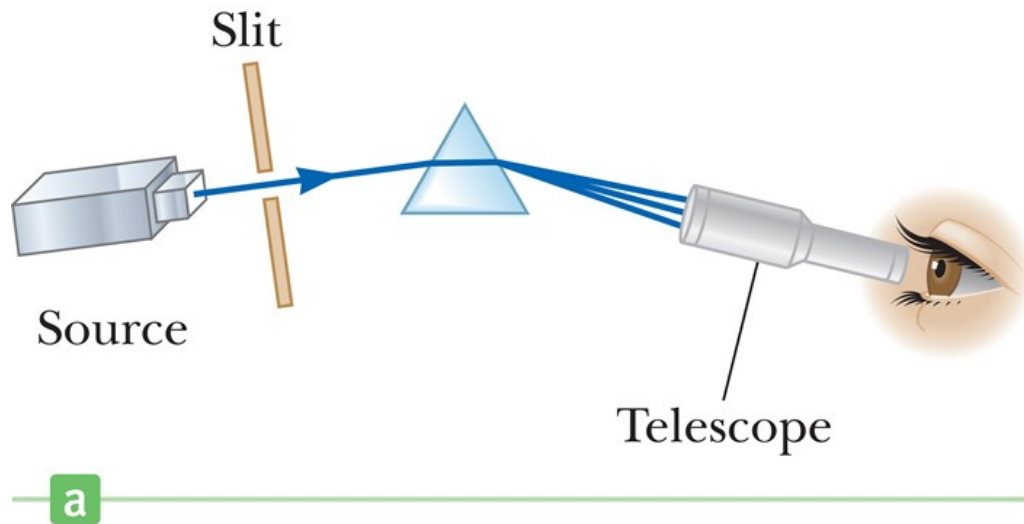
Dispersion and Prisms (3 of 5)



Different colors of light that pass through a prism are refracted at different angles because the index of refraction of the glass depends on wavelength. Violet light bends the most, red light the least.

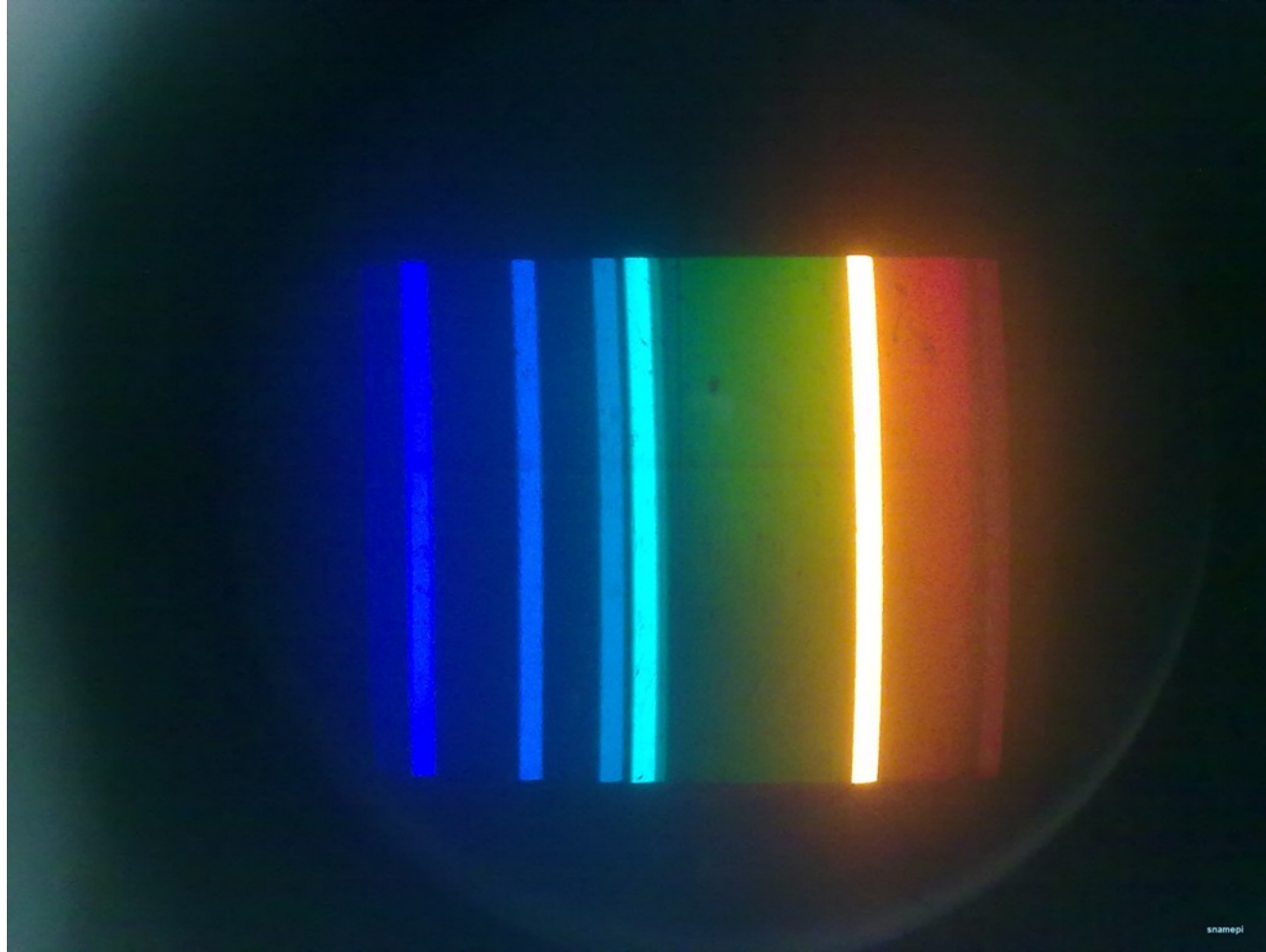


Dispersion and Prisms (4 of 5)

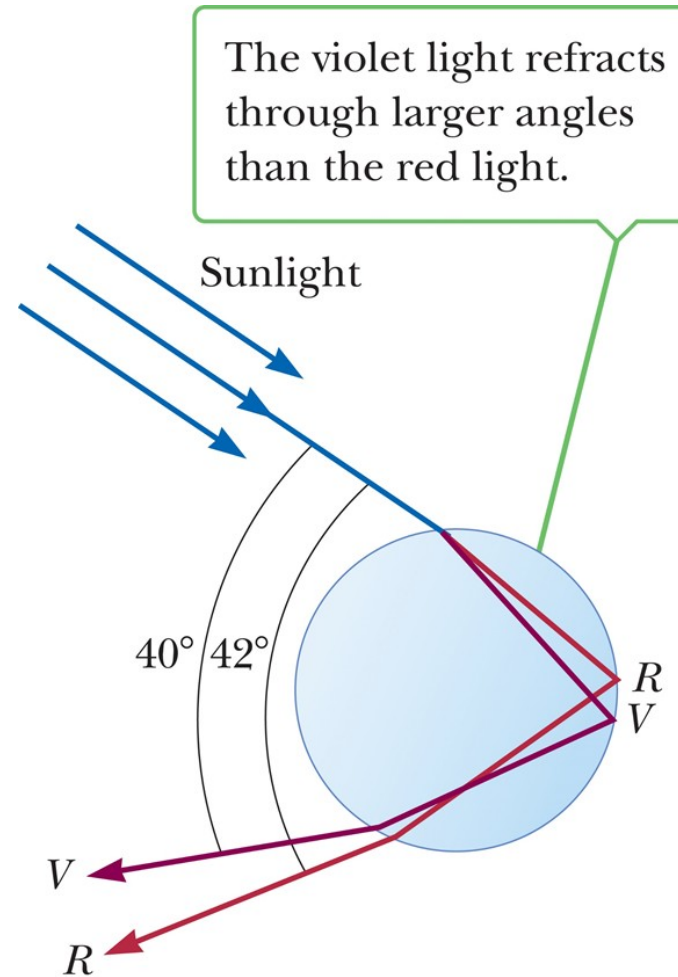


Courtesy of PASCO Scientific

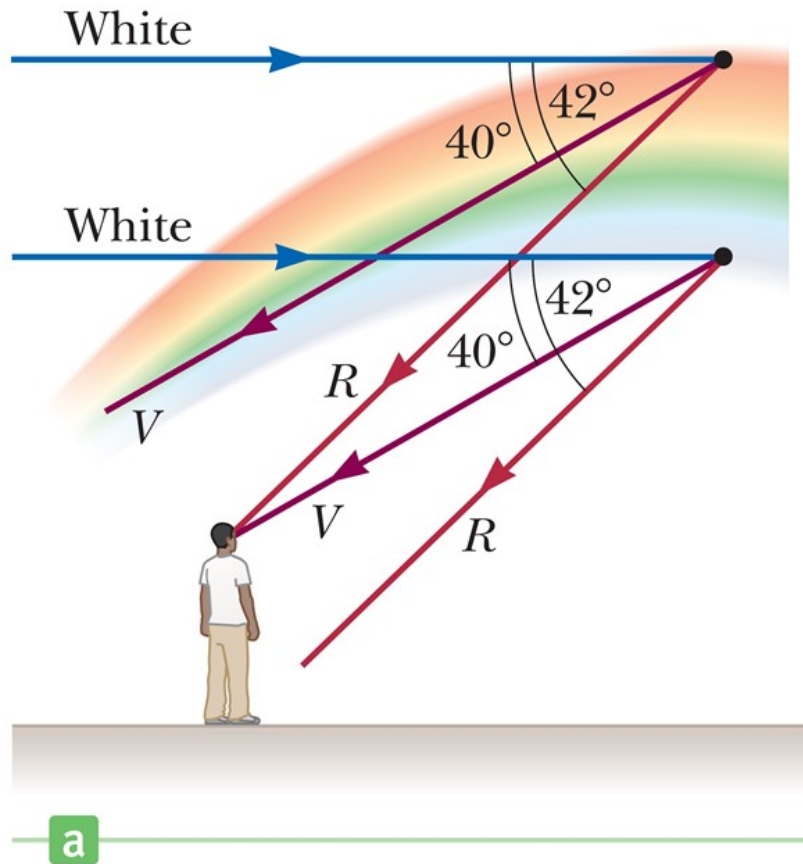
Dispersion and Prisms (5 of 5)



The Rainbow (1 of 2)



The Rainbow (2 of 2)



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